

DR. DOC TUTORIAL FOR MEDICINE STUDENTS

100 Comprehensive MCQs on Nucleic Acids

Exam Practice Booklet: This compilation contains 100 high-yield multiple-choice questions covering the structural chemistry, historical milestones, replication, transcription, translation, and biomedical applications of nucleic acids. Grounded strictly in *Harvey's Illustrated Reviews* and *Marks' Basic Medical Biochemistry*.

I. Introduction, Overview & Central Dogma

Q1. Which statement best describes the 'Central Dogma' of molecular biology as originally articulated?

- | | |
|---|--|
| A. RNA is replicated directly into protein without DNA templates | B. Genetic information flows unidirectionally from DNA to RNA to Protein |
| C. Proteins are reverse-translated into mRNA inside the nucleolus | D. DNA is directly translated into amino acids by ribosomes |

Q2. Which class of infectious agents represents an exception to the classical central dogma by using reverse transcriptase to copy RNA into DNA?

- | | |
|-----------------------------|-------------------|
| A. Adenoviruses | B. Bacteriophages |
| C. Retroviruses (e.g., HIV) | D. Prions |

Q3. In eukaryotic cells, where is the vast majority of genomic DNA localized?

- | | |
|--------------------------|-------------------------|
| A. Lysosomes | B. Nucleus as chromatin |
| C. Endoplasmic Reticulum | D. Peroxisomes |

Q4. What proportion of cellular DNA in a eukaryotic cell resides autonomously inside the mitochondria?

- | | |
|-------------------|----------------------|
| A. Less than 0.1% | B. Approximately 10% |
| C. Exactly 25% | D. Over 50% |

Q5. Which organelle organizes eukaryotic ribosomal RNA (rRNA) transcription and ribosome subunit assembly?

- | | |
|--------------------|---------------|
| A. Golgi apparatus | B. Nucleolus |
| C. Smooth ER | D. Centrosome |

Q6. Prokaryotic chromosomal DNA differs from eukaryotic nuclear DNA in that prokaryotic DNA is:

- | | |
|---|---|
| A. Linear and bound to histones | B. A single circular supercoiled molecule in a nucleoid |
| C. Enclosed within a double-membrane envelope | D. Composed exclusively of single-stranded RNA |

Q7. Small, extrachromosomal, circular double-stranded DNA molecules found in bacteria that often carry antibiotic resistance genes are called:

- A. Introns
- B. Plasmids
- C. Exons
- D. Nucleosomes

Q8. In human cells, the transcription of a gene produces an initial linear RNA copy containing both coding and noncoding regions called:

- A. Mature mRNA
- B. Primary transcript / pre-mRNA / hnRNA
- C. Transfer RNA (tRNA)
- D. Small interfering RNA (siRNA)

Q9. Coupled transcription and translation (simultaneous mRNA synthesis and protein translation) occurs in:

- A. Human erythroblasts
- B. Eukaryotic pancreatic cells
- C. Prokaryotic bacterial cells
- D. Fungal myocytes

Q10. Which functional molecule represents the final product of gene expression for noncoding genes like snRNA and tRNA?

- A. Polypeptide chain
- B. Functional RNA molecule
- C. Monosaccharide
- D. Phospholipid

II. History of DNA Discovery

Q11. Who first isolated DNA in 1865 from pus scraped from surgical bandages, initially calling it 'nuclein'?

- A. Oswald Avery
- B. Erwin Chargaff
- C. Frederick Meischer
- D. James Watson

Q12. In 1944, landmark transformation experiments performed by which scientist(s) proved that DNA is the actual genetic material?

- A. Watson and Crick
- B. Oswald Avery, Colin MacLeod, and Maclyn McCarty
- C. Maurice Wilkins and Rosalind Franklin
- D. Robert Holley

Q13. Erwin Chargaff's rules of DNA base composition state that in any double-stranded DNA sample:

- A. %A = %G and %C = %T
- B. %A = %T and %G = %C, and Total Purines = Total Pyrimidines
- C. %A + %T always equals 90%
- D. %G + %C = 100% in all species

Q14. Which experimental technique carried out by Maurice Wilkins and Rosalind Franklin provided key physical data for Watson and Crick's double helix model?

- A. Mass spectrometry
- B. X-ray diffraction / crystallography
- C. Northern blotting
- D. Agarose gel electrophoresis

Q15. In what year did James Watson and Francis Crick publish their revolutionary ~900-word paper describing the double helical structure of B-DNA?

- A. 1865
- B. 1944
- C. 1953
- D. 1975

Q16. Who made the initial breakthrough in cracking the genetic code in 1961 by showing that synthetic poly(U) directed polyphenylalanine synthesis?

- A. Marshall Nirenberg
- B. Barbara McClintock
- C. Frederick Sanger
- D. Oswald Avery

Q17. Which nucleic acid was the first to be fully sequenced in 1965 by Robert Holley?

- A. Human Chromosome 1
- B. Alanine Transfer RNA (tRNA)
- C. Bacteriophage Lambda DNA
- D. E. coli 23S rRNA

Q18. Barbara McClintock was awarded a Nobel Prize for discovering mobile genetic elements capable of moving about the genome, known as:

- A. Plasmids
- B. Transposons ('jumping genes')
- C. Histones
- D. Okazaki fragments

III. Chemistry of Nucleotides & Nucleosides

Q19. A complete nucleotide monomer consists of which three chemical components?

- A. Fatty acid, glycerol, and amino acid
- B. Nitrogenous base, pentose sugar, and phosphate group(s)
- C. Purine base, hexose sugar, and sulfate group
- D. Pyrimidine base, amino acid, and lipid

Q20. How does a nucleoside structurally differ from a nucleotide?

- A. A nucleoside lacks a nitrogenous base
- B. A nucleoside lacks phosphate groups
- C. A nucleoside contains a hexose instead of a pentose
- D. A nucleoside is linked by peptide bonds

Q21. Purines are heterocyclic double-ring structures. Which nitrogenous bases belong to the purine family?

- A. Cytosine and Thymine
- B. Adenine and Guanine
- C. Uracil and Cytosine
- D. Thymine and Uracil

Q22. Pyrimidines are single-ring structures. Which pyrimidine base is found exclusively in DNA under normal conditions?

- A. Cytosine
- B. Uracil
- C. Thymine
- D. Guanine

Q23. What is the precise structural difference between Thymine and Uracil?

- A. Thymine possesses a methyl group at C5 of its ring, whereas Uracil lacks it
- B. Uracil has an extra hydroxyl group at C2'
- C. Thymine is a purine, whereas Uracil is a pyrimidine
- D. Uracil contains a sulfur atom

Q24. What chemical feature distinguishes 2'-deoxyribose (in DNA) from D-ribose (in RNA)?

- A. Deoxyribose lacks an oxygen atom at carbon 2' (has -H instead of -OH)
- B. Ribose is a 6-carbon hexose sugar
- C. Deoxyribose contains an amino group at carbon 3'
- D. Ribose lacks an anomeric carbon

Q25. In a nucleoside, the nitrogenous base is covalently attached to carbon 1' of the pentose sugar via what type of linkage?

- A. 3'->5' phosphodiester bond
- B. β -N-Glycosidic bond
- C. Peptide bond
- D. Phosphoanhydride bond

Q26. Why are the carbon atoms of the pentose sugar in nucleosides numbered with primes (1', 2', 3', 4', 5')?

- A. To indicate they are in the D-conformation
- B. To distinguish sugar carbons from the unprimed integer numbers of the base ring
- C. To show they carry negative charges
- D. To designate optical rotation

Q27. To which carbon atom of the pentose sugar is the primary phosphate group attached in a nucleoside monophosphate?

- A. Carbon 1'
- B. Carbon 2'
- C. Carbon 3'
- D. Carbon 5'

Q28. In ATP, the linkages connecting the second (β) and third (γ) phosphate groups are termed:

- A. Phosphodiester bonds
- B. High-energy phosphoanhydride bonds
- C. Hydrogen bonds
- D. Glycosidic bonds

Q29. What property causes DNA and RNA polymers to carry net negative charges and act as polyanionic acids at physiological pH?

- A. Hydrophobic purine bases
- B. Dissociated hydroxyl groups on phosphate backbones
- C. Ribose 2'-OH groups
- D. Basic amino acid side chains

Q30. What is the standard ribonucleoside name for Adenine + Ribose?

- A. Adenylate
- B. Adenosine
- C. Deoxyadenosine
- D. Adenine monophosphate

Q31. What is the standard ribonucleoside name for Cytosine + Ribose?

- A. Cytidylate
- B. Cytidine
- C. Deoxycytidine
- D. Cytosine monophosphate

Q32. Which nucleotide serves as an activated donor of glucosyl residues in glycogen synthesis?

- A. CDP-choline
- B. UDP-glucose
- C. ATP
- D. S-adenosylmethionine

Q33. Which nucleotide acts as an activated carrier in phospholipid biosynthesis?

- A. CDP-choline
- B. UDP-galactose
- C. cGMP
- D. dGTP

Q34. Which second messenger is synthesized from ATP by the enzyme adenylyl cyclase in signal transduction?

- A. cGMP
- B. Cyclic AMP (cAMP)
- C. Inositol trisphosphate
- D. PRPP

Q35. Unusual modified bases such as dihydrouracil and pseudouridine are most abundantly concentrated in which class of RNA?

- A. Messenger RNA (mRNA)
- B. Transfer RNA (tRNA)
- C. 28S Ribosomal RNA
- D. Small nuclear RNA

IV. Synthetic Analogues & Clinical Pharmacology

Q36. Zidovudine (AZT), an antiviral drug used in HIV treatment, acts as an analogue of which natural nucleoside?

- A. Deoxyadenosine
- B. Deoxythymidine (Thymidine)
- C. Cytidine
- D. Guanosine

Q37. By what mechanism do nucleoside analogues like AZT and didanosine (ddI) terminate DNA strand elongation?

- A. They cleave the template strand
- B. They lack a free 3'-hydroxyl (-OH) group needed to form the next 3'→5' phosphodiester bond
- C. They inhibit RNA polymerase I
- D. They bind directly to histones

Q38. Cytarabine (cytosine arabinoside, araC) is an anticancer chemotherapeutic drug in which the ribose sugar is replaced by:

- A. Fructose
- B. Arabinose
- C. Deoxyribose
- D. Glucose

Q39. The antineoplastic drug 5-Fluorouracil (5-FU) is metabolically converted to 5-FdUMP to directly inhibit which enzyme?

- A. Dihydrofolate reductase
- B. Thymidylate synthase
- C. Xanthine oxidase
- D. PRPP synthetase

Q40. Which anticancer and immunosuppressive drug competitively inhibits dihydrofolate reductase (DHFR), preventing THF regeneration?

- A. Methotrexate
- B. Allopurinol
- C. Hydroxyurea
- D. Dactinomycin

Q41. How do drugs like doxorubicin (Adriamycin) and dactinomycin (actinomycin D) interfere with nucleic acid function?

- A. They degrade ribosomes
- B. They intercalate (slip) between stacked base pairs of DNA, inhibiting replication and transcription
- C. They remove poly-A tails
- D. They inhibit CTP synthetase

Q42. The chemotherapy drug hydroxyurea inhibits DNA synthesis by destroying the free radical required for activity of:

- A. Ribonucleotide reductase
- B. Xanthine oxidase
- C. Adenosine deaminase
- D. DNA ligase

Q43. Plant methylxanthines like Caffeine (coffee), Theophylline (tea), and Theobromine (cocoa) prolong cAMP action by inhibiting:

- A. Adenylyl cyclase
- B. cAMP phosphodiesterase
- C. Protein kinase A
- D. Urate oxidase

Q44. Allopurinol is used in the clinical management of gout because it directly inhibits which purine-degrading enzyme?

- A. Adenosine deaminase
- B. Xanthine oxidase
- C. HGPRT
- D. PRPP amidotransferase

Q45. In severe combined immunodeficiency (SCID) caused by adenosine deaminase (ADA) deficiency, toxic accumulation of which metabolite inhibits ribonucleotide reductase?

- A. Uric acid
- B. dATP
- C. dCTP
- D. Orotic acid

V. Structure & Properties of DNA

Q46. In a polynucleotide chain of DNA, adjacent nucleotides are covalently joined by phosphodiester bonds connecting:

- A. 5'-hydroxyl of one sugar to 3'-hydroxyl of the next sugar
- B. 3'-hydroxyl of one sugar to 5'-hydroxyl of the adjacent sugar through a phosphate
- C. Base to base directly
- D. C1' to C2' of the same sugar

Q47. By convention, the sequence of nucleotide bases along a nucleic acid chain is always written and read in which direction?

- A. 3' to 5'
- B. 5' to 3'
- C. N-terminus to C-terminus
- D. Random

Q48. The two complementary strands of the DNA double helix run in opposite directions, a property described as:

- A. Semiconservative
- B. Antiparallel
- C. Redundant
- D. Degenerate

Q49. In Watson-Crick base pairing within double-stranded DNA, Adenine pairs with Thymine via how many hydrogen bonds?

- A. One
- B. Two
- C. Three
- D. Four

Q50. Guanine complementary pairs with Cytosine via how many hydrogen bonds in double-stranded DNA?

- A. One
- B. Two
- C. Three
- D. Four

Q51. Why does a DNA fragment with a higher G-C content possess a higher melting temperature (T_m)?

- A. G-C pairs contain 2'-OH groups
- B. G-C base pairs are held together by three hydrogen bonds, requiring more thermal energy to disrupt
- C. G-C pairs are smaller in size
- D. G-C pairs contain methyl groups

Q52. Why is double-stranded DNA completely resistant to alkaline cleavage, whereas RNA is rapidly hydrolyzed by alkali?

- A. DNA contains thymine instead of uracil
- B. DNA lacks a 2'-hydroxyl (-OH) group on its deoxyribose sugar
- C. DNA strands are antiparallel
- D. DNA is bound to histones

Q53. Treatment of double-stranded DNA with alkali results in:

- A. Cleavage of phosphodiester bonds into mononucleotides
- B. Separation of the two strands (denaturation) without breaking phosphodiester bonds
- C. Methylation of adenine residues
- D. Conversion of DNA into RNA

Q54. The loss of helical structure when dsDNA is heated or exposed to extreme pH is monitored by measuring UV light absorbance at what wavelength?

- A. 210 nm
- B. 260 nm
- C. 340 nm
- D. 540 nm

Q55. The process where separated complementary single strands of DNA re-align and re-form a double helix upon slow cooling is called:

- A. Denaturation
- B. Renaturation / Reannealing / Hybridization
- C. Splicing
- D. Translation

Q56. If one strand of a double-stranded DNA molecule contains 20% A, 25% T, 30% G, and 25% C, what is the % A in the complementary strand?

- A. 20%
- B. 25%
- C. 30%
- D. 50%

Q57. Which structural form of DNA represents the classical right-handed helix described by Watson and Crick with ~10 bp per turn?

- A. A-DNA
- B. B-DNA
- C. Z-DNA
- D. Triple-stranded DNA

Q58. A-DNA is produced by moderately dehydrating B-DNA. It is a right-handed helix characterized by:

- A. 12 bp per turn and left-handed coiling
- B. 11 bp per turn with base pairs tilted 20 degrees away from the helical axis
- C. Complete loss of purine bases
- D. Single-stranded linear structure

Q59. Z-DNA is a left-handed helix with 12 bp per turn. What structural feature gives Z-DNA its name?

- A. Base pairs arranged in circular zones
- B. Phosphate backbone that 'zigzags' along the periphery
- C. Zinc atom bound at every turn
- D. Zero hydrogen bonds between strands

Q60. The spatial twisting of the two strands in the B-DNA double helix creates major and minor grooves. What is the primary function of these grooves?

- A. Allow entry of sodium ions
- B. Provide binding access for regulatory proteins to recognize specific base sequences
- C. Permit ribose cleavage
- D. Store excess ATP

Q61. What basic, positively charged proteins are complexed with eukaryotic DNA in equal weight to form chromatin?

- A. Albumins
- B. Histones
- C. Immunoglobulins
- D. Cytokeratins

Q62. Which amino acids are present in high concentrations in histones, conferring their strong positive charges at physiological pH?

- A. Aspartate and Glutamate
- B. Arginine and Lysine
- C. Valine and Leucine
- D. Tryptophan and Tyrosine

Q63. A nucleosome core particle consists of approximately 140-147 bp of double-stranded DNA wrapped around an octamer composed of two molecules each of:

- A. H1, H2A, H2B, H3
- B. H2A, H2B, H3, and H4
- C. H1, H3, H4, and H5
- D. Protamine 1, 2, 3, and 4

Q64. Which histone species binds to the linker DNA region joining adjacent nucleosome core particles?

- A. Histone H1
- B. Histone H2A
- C. Histone H3
- D. Histone H4

Q65. Telomeres located at the terminal ends of linear eukaryotic chromosomes consist of thousands of tandem repeats of which hexameric sequence in humans?

- A. 5'-TATAAB-3'
- B. 5'-AG3T2-3' (or 5'-TTAGGG-3')
- C. 5'-CAGCAG-3'
- D. 5'-AUUAAA-3'

VI. Non-Coding DNA & Mitochondrial DNA

Q66. Approximately what percentage of the 3 billion base pairs in the human genome actually codes for functional proteins?

- A. Less than 2%
- B. About 25%
- C. Over 50%
- D. Exactly 98%

Q67. Hypermethylation of cytosine residues in 'CpG islands' located near gene promoters in human DNA generally results in:

- A. Gene activation
- B. Epigenetic gene silencing / transcriptional repression
- C. DNA strand cleavage
- D. Increased polyadenylation

Q68. Noncoding intervening sequences within eukaryotic pre-mRNAs that are spliced out prior to translation are called:

- A. Exons
- B. Introns
- C. Promoters
- D. Enhancers

Q69. Highly repetitive short tandem repeat (STR) polymorphisms in noncoding DNA regions are widely used in forensic science for:

- A. Protein structure prediction
- B. DNA fingerprinting and paternity testing
- C. Western blotting
- D. Determining viral titers

Q70. Alu sequences are repetitive DNA elements that make up 6% to 8% of the human genome. Abnormal homologous recombination between Alu repeats in the LDL-receptor gene can cause:

- A. Sickle cell anemia
- B. Familial hypercholesterolemia
- C. Gout
- D. SCID

Q71. Human mitochondrial DNA (mtDNA) is structurally organized as:

- A. 46 linear chromosomes bound to histones
- B. Closed, circular double-stranded DNA molecules
- C. Single-stranded RNA loops
- D. Triple-stranded linear helices

Q72. In eukaryotic cell division, why do germ cells, stem cells, and cancer cells avoid progressive telomere shortening and senescence?

- A. They lack DNA polymerase I
- B. They express active telomerase, an RNA-directed DNA polymerase that maintains telomere length
- C. They have circular chromosomes
- D. They do not undergo S phase

Q73. Mitochondrial DNA and its protein-synthesizing system closely resemble those of:

- A. Eukaryotic nuclear systems
- B. Bacteria (prokaryotes)
- C. Retroviruses
- D. Yeasts

Q74. Which enzyme replicates mitochondrial DNA in eukaryotic cells?

- A. DNA Polymerase α
- B. DNA Polymerase γ
- C. DNA Polymerase δ
- D. DNA Polymerase ϵ

Q75. What enzyme activity intrinsic to retroviral reverse transcriptase degrades the RNA strand of an RNA-DNA hybrid during cDNA synthesis?

- A. RNase H activity
- B. 3'->5' exonuclease proofreading
- C. Primase activity
- D. Topoisomerase type II

VII. Chemistry & Structure of RNA

Q76. RNA polymers consist of ribonucleoside monophosphates linked covalently by:

- A. 2'->3' peptide bonds
- B. 3'->5' phosphodiester bonds
- C. α -1,4-glycosidic bonds
- D. Disulfide bridges

Q77. The four standard nitrogenous bases present in RNA are:

- A. Adenine, Guanine, Cytosine, and Thymine
- B. Adenine, Guanine, Cytosine, and Uracil
- C. Adenine, Hypoxanthine, Cytosine, and Thymine
- D. Guanine, Cytosine, Uracil, and Xanthine

Q78. Primary (1°) structure of RNA refers to:

- A. The 3D globular folding
- B. The linear sequence of ribonucleotides read from 5' to 3'
- C. Hairpin loop coiling
- D. Association with ribosomal proteins

Q79. Secondary (2°) structure in single-stranded RNA molecules arises primarily from:

- A. Covalent peptide bonding
- B. Intrachain complementary base pairing (hairpins and stem-loops)
- C. Histone core wrapping
- D. Polyadenylation

Q80. In RNA secondary structures, Adenine pairs with Uracil via how many hydrogen bonds?

- A. One
- B. Two
- C. Three
- D. Four

Q81. The tertiary (3°) structure of RNA molecules (such as the compact L-shape of tRNA) is stabilized by:

- A. Hydrophobic base-stacking interactions and tertiary hydrogen bonds
- B. Disulfide bonds
- C. 5'→5' triphosphate caps
- D. Ionic salt bridges with K⁺ only

Q82. Which class of RNA constitutes the vast majority (~80%) of total cellular RNA in a typical cell?

- A. Messenger RNA (mRNA)
- B. Transfer RNA (tRNA)
- C. Ribosomal RNA (rRNA)
- D. Small nuclear RNA (snRNA)

Q83. Messenger RNA (mRNA) accounts for approximately what percentage of total cellular RNA?

- A. ~5%
- B. ~15%
- C. ~50%
- D. ~80%

Q84. Eukaryotic messenger RNA is described as 'monocistronic' because it:

- A. Contains information derived from multiple bacterial operons
- B. Carries coding information for only a single polypeptide chain
- C. Lacks a start codon
- D. Is synthesized by RNA Polymerase I

Q85. What special modification is present at the 5'-end of mature eukaryotic mRNA?

- A. A long poly-A tail
- B. A 7-methylguanosine cap attached via an unusual 5'→5' triphosphate linkage
- C. A 3'-CCA sequence
- D. A dihydrouracil loop

VIII. Major Translational Classes of RNA

Q86. What is the primary physiological function of the 5'-7-methylguanosine cap on eukaryotic mRNA?

- A. Directs splicing of exons
- B. Protects mRNA from 5'→3' exonuclease attack and aids ribosome recognition for translation initiation
- C. Acts as a polyadenylation signal
- D. Codes for methionine

VII. Major Translational Classes of RNA

Q87. A sequence of 40 to 200+ adenine residues added post-transcriptionally to the 3'-end of eukaryotic pre-mRNA is called the:

- A. 5'-Cap
- B. Poly-A tail
- C. Shine-Dalgarno sequence
- D. TATA box

VIII. Major Translational Classes of RNA

Q88. Consensus signal sequence AAUAAA located near the 3'-end of eukaryotic pre-mRNA specifies:

- A. Transcription initiation site
- B. Cleavage and polyadenylation site for poly-A tail addition
- C. Splice donor site
- D. Binding site for RNA Polymerase I

Q89. Transfer RNA (tRNA) molecules serve as 'adaptor' molecules during translation. What functional sequence is located at the unpaired 3'-end of every mature tRNA?

- A. 5'-AUG-3'
- B. 3'-CCA-5' (or 5'-CCA-3' terminal hydroxyl)
- C. 5'-UAA-3'
- D. Poly-A tail

Q90. During tRNA charging, aminoacyl-tRNA synthetase attaches the amino acid to the 3'-CCA end of tRNA via a bond between:

- A. α -carboxyl group of amino acid and 3'-OH of terminal adenosine
- B. α -amino group of amino acid and phosphate group
- C. Side chain and anticodon loop
- D. Methyl group and ribose C2'

Q91. The three-nucleotide sequence located on the anticodon loop of tRNA that complementary base-pairs with an mRNA codon is called the:

- A. Codon
- B. Anticodon
- C. Promoter
- D. Cistron

Q92. Which functional loop of tRNA contains dihydrouracil and serves as a major recognition site for aminoacyl-tRNA synthetase?

- A. Anticodon arm
- B. D-arm / D-loop
- C. T Ψ C arm
- D. Acceptor stem

Q93. What are the sedimentation coefficients of the four rRNA species present in eukaryotic 80S cytosolic ribosomes?

- A. 5S, 16S, 23S
- B. 5S, 5.8S, 18S, and 28S
- C. 4S, 10S, 30S, 50S
- D. 7S, 12S, 14S, 21S

Q94. Prokaryotic 70S ribosomes contain which three specific rRNA species?

- A. 5S, 16S, and 23S
- B. 5S, 5.8S, and 28S
- C. 18S, 28S, and 4S
- D. 16S, 18S, and 23S

Q95. Which rRNA molecule in the large ribosomal subunit possesses intrinsic 'peptidyl transferase' ribozyme activity that catalyzes peptide bond formation?

- A. 16S rRNA (prokaryotes) / 18S rRNA (eukaryotes)
- B. 23S rRNA (prokaryotes) / 28S rRNA (eukaryotes)
- C. 5S rRNA
- D. 5.8S rRNA

IX. Small Noncoding RNAs & RNAi

Q96. Small nuclear RNAs (snRNAs) combine with specific nuclear proteins to form snRNPs ('snurps'), which assemble to construct the:

- A. Ribosome
- B. Spliceosome
- C. Proteasome
- D. Nucleosome

Q97. Small nucleolar RNAs (snoRNAs) reside in the nucleolus to perform what specific biological role?

- A. Polyadenylation of mRNA
- B. Directing cleavage, trimming, and base/ribose modifications of pre-rRNA precursors
- C. Reverse transcription
- D. Translation termination

Q98. MicroRNAs (miRNAs) are endogenous ~22 bp noncoding RNAs that regulate gene expression by associating with which cytosolic complex?

- A. Spliceosome
- B. RNA-induced Silencing Complex (RISC)
- C. DNA Polymerase III holoenzyme
- D. CAP-cAMP complex

Q99. Which endonuclease enzyme processes precursor double-stranded RNAs in the cytoplasm to generate mature miRNAs and siRNAs?

- A. Dicer
- B. DNA Ligase
- C. PRPP Synthetase
- D. Xanthine Oxidase

Q100. In RNA interference (RNAi), if a miRNA or siRNA guide strand forms PERFECT base-pairing complementarity with its target mRNA, the outcome is:

- A. Inhibition of splicing only
- B. Direct endonucleolytic cleavage/degradation of the target mRNA by Argonaute (Slicer)
- C. Stimulation of translation
- D. Poly-A tail extension

COMPLETE ANSWER KEY & EXPLANATIONS

Dr. Doc Tutorial - Verified Medical Biochemistry Reference

Q1: [B] - The Central Dogma states that DNA replicates to preserve information, is transcribed into RNA, and RNA is translated into protein.

Q2: [C] - Retroviruses carry an RNA genome and use viral reverse transcriptase to make a cDNA copy that integrates into host DNA.

Q3: [B] - Eukaryotic genomic DNA is linear and housed in the nucleus as chromatin (DNA bound to histones).

Q4: [A] - Mitochondrial DNA (mtDNA) constitutes less than 0.1% of total cellular DNA (<20,000 bp).

Q5: [B] - The nucleolus is a substructure of the nucleus dedicated to rRNA transcription, processing, and ribosome assembly.

Q6: [B] - Prokaryotes lack a nuclear membrane and contain a single, circular, supercoiled chromosome in a region called the nucleoid.

Q7: [B] - Plasmids are autonomous circular DNA elements in bacteria that can carry antibiotic resistance traits.

Q8: [B] - Primary transcripts (hnRNA or pre-mRNA) contain exons and introns before co- and post-transcriptional splicing.

Q9: [C] - Because bacteria lack a membrane-bound nucleus, ribosomes attach to mRNA while it is still being transcribed.

Q10: [B] - For noncoding genes, the final functional product of gene expression is the RNA molecule itself rather than a protein.

Q11: [C] - Frederick Meischer isolated DNA in 1865 from white blood cells in pus on surgical bandages.

Q12: [B] - Avery, MacLeod, and McCarty demonstrated in 1944 that DNA was the transforming principle of heredity.

Q13: [B] - Chargaff established that A = T, G = C, and the sum of purines (A+G) equals pyrimidines (C+T).

Q14: [B] - X-ray diffraction images of DNA fibers taken by Franklin and Wilkins revealed the helical parameters.

Q15: [C] - Watson and Crick published their structural model of B-DNA in Nature in 1953.

Q16: [A] - Nirenberg demonstrated that poly(U) mRNA synthesized polyphenylalanine, proving UUU codes for Phenylalanine.

Q17: [B] - Robert Holley deduced the complete nucleotide sequence of yeast alanine tRNA in 1965.

Q18: [B] - McClintock discovered transposable genetic elements (transposons) in maize.

Q19: [B] - A nucleotide is composed of a nitrogenous base, a pentose monosaccharide, and 1-3 phosphate groups.

Q20: [B] - Nucleoside = Base + Pentose sugar. Nucleotide = Nucleoside + Phosphate group(s).

Q21: [B] - Adenine (A) and Guanine (G) are double-ring purines present in both DNA and RNA.

Q22: [C] - Thymine (5-methyluracil) is unique to DNA, whereas Uracil replaces thymine in RNA.

Q23: [A] - Thymine differs from uracil only by possessing a methyl group at position 5 of its pyrimidine ring.

Q24: [A] - 2'-Deoxyribose lacks a hydroxyl group at position C2', carrying hydrogen (-H) instead.

Q25: [B] - The base attaches to N9 (purines) or N1 (pyrimidines) to C1' of sugar via a β -N-glycosidic bond.

Q26: [B] - Primes (') differentiate sugar ring carbons from integer-numbered ring atoms in purines/pyrimidines.
Q27: [D] - The first phosphate group forms an ester bond with the 5'-hydroxyl group of the pentose sugar.
Q28: [B] - The β and γ phosphate bonds are high-energy phosphoanhydride linkages releasing ~ 7.3 kcal/mol upon hydrolysis.
Q29: [B] - Phosphate groups dissociate protons at physiological pH, leaving negatively charged oxygens along the backbone.
Q30: [B] - Adenine + Ribose = Adenosine (purine nucleosides end in -osine).
Q31: [B] - Cytosine + Ribose = Cytidine (pyrimidine nucleosides end in -idine).
Q32: [B] - UDP-glucose is the activated carbohydrate intermediate required by glycogen synthase.
Q33: [A] - CDP-choline and CDP-diacylglycerol serve as activated intermediates in lipid biosynthesis.
Q34: [B] - Adenylyl cyclase converts ATP into cyclic AMP (cAMP) upon hormone activation.
Q35: [B] - 10% to 20% of nucleotides in tRNA undergo post-transcriptional modification to unusual bases.
Q36: [B] - AZT is an analogue of deoxythymidine carrying a 3'-azido (-N ₃) group instead of a 3'-OH.
Q37: [B] - Lacking a 3'-OH group, their incorporation prevents polymerases from adding the next incoming nucleotide.
Q38: [B] - AraC contains arabinose in place of ribose, blocking DNA replication in rapidly dividing cancer cells.
Q39: [B] - 5-FdUMP acts as a suicide inhibitor of thymidylate synthase, stopping dTMP synthesis.
Q40: [A] - Methotrexate inhibits DHFR, blocking thymidylate and purine synthesis.
Q41: [B] - Planar multiring intercalating agents insert between stacked base pairs, disrupting template function.
Q42: [A] - Hydroxyurea inhibits ribonucleotide reductase, preventing conversion of NDPs to dNDPs.
Q43: [B] - Methylxanthines competitively inhibit cAMP phosphodiesterase, keeping intracellular cAMP levels elevated.
Q44: [B] - Allopurinol inhibits xanthine oxidase, preventing uric acid formation from hypoxanthine and xanthine.
Q45: [B] - ADA deficiency leads to dATP accumulation, which strongly inhibits ribonucleotide reductase and halts immune cell proliferation.
Q46: [B] - 3'->5' phosphodiester bonds join the 3'-OH of one pentose to the 5'-OH of the next pentose via a phosphate.
Q47: [B] - Nucleic acid sequences are universally read and written from the 5'-end to the 3'-end.
Q48: [B] - Antiparallel orientation means one strand runs 5'->3' while the partner strand runs 3'->5'.
Q49: [B] - Adenine forms two hydrogen bonds with Thymine (A=T).
Q50: [C] - Guanine forms three hydrogen bonds with Cytosine (G=C).
Q51: [B] - Three H-bonds in G-C pairs make the helix more thermally stable than two H-bonds in A-T pairs.
Q52: [B] - Lacking a 2'-OH group, DNA cannot undergo the intramolecular nucleophilic attack that cleaves RNA in alkali.
Q53: [B] - Alkali disrupts H-bonds to separate dsDNA into single strands while leaving covalent phosphodiester bonds intact.
Q54: [B] - Purine and pyrimidine bases absorb UV light strongly at 260 nm; ssDNA exhibits higher absorbance (hyperchromicity) than dsDNA.
Q55: [B] - Renaturation or hybridization occurs when complementary single strands re-form hydrogen bonds.

Q56: [B] - Adenine on the complementary strand pairs with Thymine on the first strand (25% T -> 25% A).
Q57: [B] - B-DNA is the predominant right-handed form in physiological aqueous environments (10-10.4 bp/turn).
Q58: [B] - A-DNA has 11 bp/turn, tilted base pairs, and is adopted by DNA-RNA hybrids and dsRNA.
Q59: [B] - Z-DNA has a zigzagging phosphodiester backbone and occurs transiently in alternating purine-pyrimidine sequences.
Q60: [B] - Major and minor grooves expose edge groups of base pairs for sequence-specific protein binding.
Q61: [B] - Histones are small basic nuclear proteins rich in Arginine and Lysine that package eukaryotic DNA.
Q62: [B] - Arginine and Lysine side chains carry positive charges that bind electrostatically to negative DNA phosphates.
Q63: [B] - The histone octamer core contains two copies each of H2A, H2B, H3, and H4.
Q64: [A] - Histone H1 binds to the linker DNA between nucleosome cores to facilitate higher-order coiling (solenoid).
Q65: [B] - Human telomeric DNA consists of repetitive AG3T2 (TTAGGG) sequences paired with complementary Cs and As.
Q66: [A] - Less than 2% (or ~1.5%) of the human genome consists of protein-coding exons.
Q67: [B] - Methylation of 5-methylcytosine in promoter CpG islands silences gene expression.
Q68: [B] - Introns are noncoding intervening regions excised by spliceosomes; exons are joined into mature mRNA.
Q69: [B] - STR polymorphisms vary in length between individuals, providing a unique molecular profile for forensics.
Q70: [B] - Homologous recombination between Alu repeats deletes part of the LDL-receptor gene, raising blood cholesterol.
Q71: [B] - mtDNA consists of closed, circular, double-stranded molecules (<20,000 bp).
Q72: [B] - Telomerase uses its intrinsic RNA template to extend the 3'-end of telomeric DNA, preventing senescence.
Q73: [B] - Mitochondrial DNA replication and protein synthesis closely resemble bacterial (prokaryotic) systems.
Q74: [B] - DNA Polymerase γ is the specialized eukaryotic polymerase responsible for replicating mtDNA.
Q75: [A] - Reverse transcriptase possesses associated RNase H activity that specifically degrades the RNA template strand.
Q76: [B] - Like DNA, RNA nucleotides are joined covalently by 3'->5' phosphodiester linkages.
Q77: [B] - RNA contains A, G, C, and Uracil (U) instead of Thymine.
Q78: [B] - Primary structure is the specific linear nucleotide sequence read in the 5'->3' direction.
Q79: [B] - Single-stranded RNA loops back on itself to form localized helical stem-loops via intrachain H-bonding.
Q80: [B] - Adenine pairs with Uracil via two hydrogen bonds (A=U).
Q81: [A] - 3D folding into compact shapes is stabilized by hydrophobic base stacking and tertiary H-bonds.
Q82: [C] - rRNA makes up ~80% of cellular RNA, tRNA makes up ~15%, and mRNA makes up ~5%.
Q83: [A] - mRNA comprises only ~5% of cellular RNA, though it is the most heterogeneous in size and sequence.
Q84: [B] - Eukaryotic mRNA is monocistronic (codes for one protein); prokaryotic mRNA is polycistronic.
Q85: [B] - The 5'-cap is a 7-methylguanosine joined 'backward' via a 5'->5' triphosphate linkage.

Q86: [B] - The 5'-cap stabilizes mRNA against 5'-exonucleases and facilitates 40S ribosome positioning.
Q87: [B] - The 3'-poly-A tail protects mRNA from 3'-exonucleases and assists nuclear export.
Q88: [B] - AAUAAA is the polyadenylation signal recognized by endonucleases prior to poly-A tail addition.
Q89: [B] - All mature tRNAs end in a 3'-CCA sequence where the specific amino acid is covalently attached.
Q90: [A] - An ester bond forms between the α -carboxyl group of the amino acid and the 3'-OH of terminal adenosine.
Q91: [B] - The anticodon on tRNA base-pairs in antiparallel orientation with the triplet codon on mRNA.
Q92: [B] - The D-arm contains dihydrouracil and mediates synthetase enzyme recognition.
Q93: [B] - Eukaryotic 80S ribosomes contain 28S, 5.8S, and 5S (in 60S subunit) plus 18S (in 40S subunit).
Q94: [A] - Prokaryotic 70S ribosomes consist of 23S and 5S (in 50S subunit) plus 16S (in 30S subunit).
Q95: [B] - 23S rRNA in bacteria and 28S rRNA in eukaryotes act as ribozymes catalyzing peptide bond synthesis.
Q96: [B] - snRNAs (U1, U2, U4, U5, U6) and proteins form snRNPs that assemble into spliceosomes to excise introns.
Q97: [B] - snoRNAs guide chemical modifications (methylation, pseudouridylation) and trimming of pre-rRNA.
Q98: [B] - Single-stranded miRNA guides the RISC complex to complementary target mRNAs for gene silencing.
Q99: [A] - Dicer cleaves cytosolic precursor dsRNAs into ~21-23 bp miRNA/siRNA duplexes.
Q100: [B] - Perfect complementarity induces Argonaute (Slicer) in RISC to cleave the target mRNA; partial match leads to translation repression.